

Jhabriel Allen

Professor Sridhar Ganti

ITAI 2372

June 26, 2026

Industry Overview

The industry I chose is AI and blockchain integration. This is not just one single industry, but more of an emerging technology space where two powerful tools are being combined to solve problems around trust, data, automation, and security. Artificial intelligence is used to analyze information, make predictions, detect patterns, and automate decision-making. Blockchain is used to create a secure, decentralized, and tamper-resistant digital record. When these two technologies work together, they can make systems smarter and more trustworthy at the same time.

This industry serves many groups, including businesses, healthcare organizations, financial institutions, supply chain companies, governments, educators, and even everyday consumers. For example, a hospital could use AI to help analyze patient data while blockchain helps protect the integrity of that data. A retail company could use AI to predict product demand while blockchain tracks where the products came from. A financial company could use AI to detect fraud while blockchain creates a transparent record of transactions.

This topic is relevant today because the world is becoming more digital, but digital systems still have major trust issues. People want to know if their data is safe. Businesses want to know if information has been changed or manipulated. Customers want transparency about the products they buy. At the same time, AI is growing fast, and many people are concerned about

whether AI systems are accurate, fair, and explainable. Blockchain does not magically solve every AI problem, but it can help create a stronger foundation for accountability, transparency, and data integrity. In simple terms, AI brings the intelligence, and blockchain brings the receipts.

Current AI Trends

One major trend in AI and blockchain integration is using blockchain to improve data integrity for AI systems. AI depends heavily on data. If the data is inaccurate, biased, outdated, or manipulated, the AI results can also be wrong. Blockchain can help by creating a record of where data came from, when it was added, and whether it has been changed. This is important because AI systems are only as good as the information they are trained on. If the data is messy, the output will be messy too. Blockchain helps create a more trusted data trail.

This is especially useful in supply chain management. Walmart is a good example because the company has used blockchain through IBM Food Trust to improve food traceability. Instead of taking days or weeks to track where a food product came from, blockchain can help trace the product back to its source in seconds. AI can then support the supply chain by helping predict demand, optimize inventory, and reduce waste. Together, the two technologies help companies respond faster to problems, protect consumers, and make better business decisions. This is one of the more practical examples because it is not just technology for the sake of sounding advanced. It solves a real problem: knowing where products came from and acting quickly when something goes wrong.

A second trend is using AI and blockchain in finance. Financial institutions already use AI to detect suspicious activity and patterns that could suggest fraud. Blockchain adds another layer by creating transparent and tamper-resistant transaction records. This can help with auditing, compliance, and trust. In this case, AI is looking for patterns, while blockchain is helping

preserve the record. Smart contracts are another part of this trend. Smart contracts can automatically execute agreements when certain conditions are met. AI can make those systems more adaptive by analyzing risk, customer behavior, or transaction patterns. This could make banking, insurance, and lending faster, but it also means companies must be careful about how automated decisions are made.

A third trend is using AI and blockchain to improve intellectual property management. IPwe is an example of a company that has used AI and blockchain to make patent management more transparent and efficient. Patent systems can be complicated because ownership, licensing, valuation, and risk are not always easy to track. IPwe's platform uses AI to analyze patent value, trends, and risks, while blockchain helps create a more transparent record of ownership and transactions. This matters because intellectual property can be extremely valuable, but it can also be difficult to manage. By combining AI and blockchain, companies can better understand what assets they own and how to use them.

Ethical Implications

One major ethical concern with AI and blockchain integration is privacy. Blockchain is known for transparency, but transparency can become a problem when sensitive personal data is involved. In healthcare, finance, education, and government, data often includes private information. If that data is stored incorrectly on a blockchain, it may be difficult or impossible to remove because blockchain records are designed to be permanent. That creates a serious issue when privacy laws or personal rights require data to be corrected or deleted.

This is where organizations have to be smart about design. Not everything needs to be stored directly on-chain. Some sensitive data may need to be stored off-chain, while the blockchain only stores proof that the data exists or has not been changed. Encryption, permissioned

blockchains, and consent management can also help. Still, the ethical issue remains: organizations must balance transparency with privacy. Just because technology can track everything does not mean it should track everything.

A second ethical concern is bias and fairness in AI systems. Blockchain can help verify where data came from, but it does not automatically make the data fair. If an AI model is trained on biased data, it can still produce biased results even if that data is stored securely on a blockchain. For example, if an AI lending tool is trained on historical financial data that reflects past discrimination, the system may continue unfair patterns. Blockchain may show a clean record of the data trail, but the AI model could still make decisions that hurt certain groups.

This is important because people sometimes treat technology as if it is neutral. It is not. AI systems reflect the data, assumptions, and decisions that humans put into them. Blockchain can support accountability by making it easier to audit the process, but people still need to examine the data, test the model, and question the results. Fairness cannot be outsourced to a machine.

A third ethical concern is job displacement. AI and blockchain together can automate tasks that used to require human workers. Smart contracts can automate agreements. AI can review documents, detect fraud, classify patents, manage inventory, and support customer service. This can make businesses more efficient, but it can also reduce the need for certain administrative, compliance, logistics, and support roles. The risk is not only that jobs disappear, but that workers are not given enough time or training to transition into new roles.

At the same time, this industry can also create new opportunities. Businesses will need people who understand AI, blockchain, cybersecurity, compliance, data governance, and digital strategy. The ethical responsibility is making sure workers are not left behind. Companies should not just adopt these tools to cut costs. They should also invest in upskilling employees so people can move into higher-value work.

Another concern is transparency and explainability. AI can already be difficult to understand, especially when models are making complex predictions. Blockchain can create an audit trail, but that does not mean the average person will understand the decision. For example, a blockchain record might show what data was used in an AI decision, but the person affected by that decision still needs a clear explanation. People deserve to understand why an AI system denied them a loan, flagged a transaction, or influenced a healthcare recommendation. Transparency should not just mean “the data is available somewhere.” It should mean people can actually understand what happened.

Future Trends

In the next five to ten years, AI and blockchain integration will likely become more common in industries where trust, data accuracy, and accountability matter. Healthcare, finance, supply chain management, education, government, and intellectual property are all strong areas for growth. The main reason is simple: AI needs reliable data, and blockchain can help prove that data has not been tampered with. As more organizations use AI for serious decisions, they will need stronger systems for tracking how those decisions were made.

One future trend is decentralized AI marketplaces. These would allow people and organizations to securely exchange AI models, datasets, or computing power. Blockchain could help manage ownership, payments, licensing, and access. This could open the door for smaller companies or independent developers to participate in AI without needing the same massive infrastructure as big tech companies. Instead of AI being controlled only by the largest organizations, blockchain could support a more open marketplace.

Another future trend is federated learning on blockchain. Federated learning allows AI models to be trained across different devices or organizations without moving all the raw data into one

central place. This is important for privacy. For example, hospitals could collaborate to improve an AI diagnostic model without sharing all patient records directly. Blockchain could help record model updates, verify participation, and make the process more trustworthy. This could be a major opportunity in healthcare, finance, and other industries that deal with sensitive data.

Edge computing is another area that could grow. Edge computing means processing data closer to where it is created, such as on devices, sensors, vehicles, or local networks. When AI works at the edge, it can make faster decisions because it does not have to send everything back to a central server. Blockchain can help secure and verify the data coming from those devices. This could support smart cities, autonomous vehicles, remote healthcare monitoring, and industrial systems.

There will also be more innovation around digital identity and content authenticity. As AI-generated content becomes more realistic, people will need better ways to verify what is real, what is edited, and where content came from. Blockchain could be used to create records of digital content ownership, source, and changes over time. This could help with misinformation, copyright protection, and online trust. With deepfakes and synthetic media becoming more common, this is not just a tech issue. It is a public trust issue.

However, there are risks. Scalability is still a challenge because some blockchain networks are slow or expensive when transaction volumes are high. Interoperability is another issue because different blockchain platforms and AI systems may not easily work together. Regulation will also play a major role. Governments will need to decide how to handle privacy, automated decision-making, digital ownership, and accountability. The opportunity is huge, but the industry will need clear standards and responsible leadership.

Reflection

What surprised me most about this topic is how practical the combination of AI and blockchain can be. Before learning more about it, I mostly connected blockchain with cryptocurrency. I did not fully realize how useful it could be in areas like healthcare, supply chain management, finance, education, and intellectual property. The idea that blockchain can help prove the quality and history of data makes a lot of sense, especially as AI becomes more involved in decision-making.

I also found it interesting that blockchain can help address some of the trust issues people have with AI. A lot of people are concerned about AI being a “black box,” where no one really knows how a decision was made. Blockchain does not solve that problem by itself, but it can help create a better audit trail.

I would be interested in working in this industry, but probably not as a blockchain developer. I would be more interested in the strategy, education, and implementation side. I could see myself helping organizations understand how AI and blockchain could support better workflows, safer data practices, and stronger decision-making. A lot of companies hear terms like AI and blockchain and immediately get overwhelmed. I would want to help translate the technology into practical use cases that actually make sense for the organization.

Overall, I think this industry has a lot of potential, but it needs balance. AI and blockchain together can create more secure, transparent, and efficient systems. The real value of the technology will come from how responsibly people choose to use it.

Citations

Clarivate. (2022). *Clarivate partners with IPwe to enhance AI and blockchain patent solutions.*

Ethereum Foundation. (n.d.). *Ethereum energy consumption.*

Houston Community College. (2024). *ITAI 2372 Mod 08 Blockchain Integration Final* [PowerPoint slides].

IBM. (n.d.). *AI and blockchain help discover and transact IP*.

LF Decentralized Trust. (n.d.). *How Walmart brought unprecedented transparency to the food supply chain with Hyperledger Fabric*.

Tsolakis, N., Niedenzu, D., Simonetto, M., Dora, M., & Kumar, M. (2022). Artificial intelligence and blockchain implementation in supply chains: A pathway to sustainability and data monetisation? *Annals of Operations Research*.